**Lesson 4.1** **Duration: 40 minutes**

# Introduction to Data

## Learning Objectives

- Define data and distinguish between data and information
- Identify different types of data in daily life
- Explain why computers represent data as numbers

## Engage (5-7 minutes)

Think of a farmer counting sacks of grain—each number is raw data. When he adds them up and says "I have 50 sacks worth ₹25,000," that becomes useful information. Today we learn what data is and how computers handle it.

## Explore (10-12 minutes)

Look around the classroom and list 10 pieces of raw data—names, roll numbers, heights, dates, temperatures. Sort them into categories: numbers, text, and dates. Discuss with a partner which ones become information when combined.

## Explain (10-12 minutes)

Data is raw, unprocessed facts—numbers, letters, or symbols. Information is processed data that has meaning. Computers store everything as numbers because their circuits understand only on and off states. Text, images, and sounds are all converted to numbers internally.

## Elaborate (8-10 minutes)

Your school attendance register contains data—names, dates, present/absent marks. Write how this raw data becomes information: "Class 6 had 95% attendance this month." Identify three more examples where data becomes information in daily life.

## Evaluate (5-8 minutes)

Exit ticket: Define data and information with one example each. Explain in one sentence why computers use numbers to represent all types of data.